Alicia Garrido Peña, Ph.D.

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Personal details

Name	Alicia Garrido Peña
Current activity	Assistant Professor in Computer Engineering. Research member of Biological Neurocomputational Group (GNB)
Nationality	Spain
Affiliation	Departamento de Ingeniería Informática. Universidad Autónoma de Madrid.

Academic Background

2014 – 2018	**Computer Science Engineering** University of Granada.
2018 – 2019	**M.Sc. Computer Science Engineering** Autonomous University of Madrid (UAM).
	M.Sc. Research and Development in Information Technologies. UAM.
2020 – 2024	**PhD in Computer Engineering and Telecomunications.** ***Cum laude*****. International mention** UAM.

Internacionalization

September - December, 2021	**Research Stay**, University of Sussex, Brighton, United Kingdom. Professor: Thomas Nowotny.
December 1st-9th, 2025	**Teaching Stay**, Universidad Politécnica Salesiana, Quito y Cuenca, Ecuador.

Skills

Electrophysiology	Dynamic clamp, intracellular (sharp electrodes) and extracellular recordings. Expertise experience in *Lymnaea stagnalis* preparation and recording and *Carcinus maenas* STG recording.
Optical neurotechnology	Expertise in Infrared Laser stimulation.
Computational neuroscience	Conductance-based models, single cells and small circuits. High experience implementing and simulating.
Real-time and closed-loop	Design and performance of experiments with hard real-time combining living and simulated neurons.
Data analysis	High experience in data analysis with Python. Basic formation in fMRi with FSL
Coding	Python, C++, Java, Kotlin, SQL, XML/XSL, PHP, LaTeX
Languages	Strong Spanish (native) and English. Basic German.
Dissemination and Science Communication	Experience in oral presentations in conferences. High interest in science communication and dissemination.

Research Publications

Courses and activities

Neuroscience

February 2024 — **fMRi Postpfrocessing Course: Resting State & FSL**, Organized by the Center for Clinical Neuroscience. Hospital los Madroños. **Hours**: 12 hours.

October 2021 — **Hackathon Brain Code Games**, Organized by the Spanish Society of Neuroscience (SENC) and the Universidad Autónoma de Madrid (UAM). **Hours**: 60 hours.

October 2020 — **First Neutouch Summer School: Touch for Prosthetics**, Online. Organizer: Neutouch.

April 2020 — **BCI & Neurotechnology Spring School 2020**, Online. Organizer: g.tec.

June 18, 2019 — **Biological Risk (4h)**, Madrid. Organizer: Prevention of Risks Service, Universidad Autónoma de Madrid (UAM).

July 4, 2019 — **Security on Chemicals Handling (4h)**, Madrid. Organizer: Prevention of Risks Service, Universidad Autónoma de Madrid (UAM).

Science disemination and interdisciplinarity

March, 2024 and 2025 — **Organization and Conduction of course titled: "Free Software for Research Carreer"**, Universidad Autónoma de Madrid (UAM).

January 25, 2023 — **Conference "Open Science in the Spanish University and its Impact on Research Development and Evaluation"**, Interuniversity Institute INAECU.

May 2021 — **#HiloTesis Dissemination Contest**, Organized by CRUE.

October 2020 to February 2021 — **Interdisciplinary Views in Scientific Research (5 ECTS)**, Madrid. Organizer: EDUAM, Universidad Autónoma de Madrid (UAM).

February to June 2018 — **Scientific Entertainer Course at Science Museum (100h; 4 ECTS)**, Granada. Organizer: Parque de las Ciencias; University of Granada (UGR).

Women in Science and STEM

June 2024 — **AI Workshop for campus "I want to be an woman engineer"**, Universidad Autónoma de Madrid (UAM).

February 2024 — **Gender-Perspective Science Communication Course**, Science Culture Unit, Universidad Autónoma de Madrid (UAM).

Courses and activities (continued)

June 2022 — **Participation in the Roundtable "Studying STEM"**, Project: "I Want to Be an Engineer - UAM", Organized by the School of Engineering. Universidad Autónoma de Madrid (UAM).

Teacher training courses

May 9–11, 2022 — **Towards an Inclusive University: Transformative Elements**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: — **Hours**: 6.

February 15–17, 2023 — **How to Evaluate Learning Outcomes**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

April 20–27, 2023 — **Detection and Management of Psychological Issues Among Students**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

October 17–24, 2023 — **The Concept of Competence and Competency-based Training**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

February 19–20, 2024 — **Authority and Leadership Course**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

May–June 20–10, 2024 — **Design and Planning of the Teaching-Learning Process in the Context of a Course. Preparation of the Course Syllabus**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 2 **Hours**: 16.

October 18–25, 2024 — **Gamification and Game-based Learning: A Practical Workshop for Creating Educational Video Games**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

October 10–23, 2024 — **Flipping the Classroom: An Introduction to the Flipped Classroom**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

May 26–29, 2025 — **Evaluating Learning**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 12.

May–June 13–04, 2025 — **Creating a Dynamic Classroom**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 12.

October 10, 2025 — **Workshop "Teaching Blindly" (Fourth Edition)**, Faculty of Psychology, Universidad Autónoma de Madrid (UAM). **Event:** Accessible Psychology: "Without Groping in the Dark." **Hours**: 2.

Courses and activities (continued)

October-November 24-7 — **Online Teaching and Intellectual Property: Practical Aspects for University Faculty**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 9.

November 4-11 — **Motivation and University Learning**, Teacher Training Program. Universidad Autónoma de Madrid (UAM). **Credits**: 1 **Hours**: 8.

Other courses

February 2021 - June 2021 — **Organization and Participation in Research Group Seminars**, Organized by GNB.

June 2023 — **Participation in the Poster Exhibition at the Doctoral Week 2023**, Organized by the Graduate School. UAM.

September 2023 — **Attendance at the International Congress IBRO**, Organized by the International Brain Research Organization.

Academic Period 2017/2018 — **University of Granada Library: Formative Actions (3 ETCS)**, Granada. Organizer: University of Granada (UGR).

Grants and other merits

December 2025 — **Erasmus+ KA-171**, Grant for teaching mobility.

2025 — **Travel Award CNS*2025 conference**, Organization for Computational Neuroscience.

Travel Award SENC 2025 conference, Sociedad Española de Neurociencia.

June 2022 — **Thesis in 3 minutes Competition (honorable mention)**, Universidad Autónoma de Madrid, Alcalá de Henares, Politécnica de Madrid, Rey Juan Carlos, and Universidad Complutense.

2018-2019 — **Competitive grant for master studies in Universidad Autonoma de Madrid**, Master research program. Universidad Autónoma de Madrid.

2016-2017 — **Erasmus+**, University of Granada.

2016 — **Best Academic Certificate Award from E.T.S. de Ingenierías Informática y de Telecomunicación**, University of Granada.

Teaching experience

2020 to the present

- Teaching experience in Computer Science Engineering, Telecommunication Engineering and Biomedical Engineering, and Master in Deep Learning for Audio and Video Processing. Courses related to Basic Programming, Operating Systems, Automata and Languages, Android programming, Deep Learning Basics.

Courses Taught

Academic Year: 2019–2020

- **Course:** Programming II
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 30h (1 lab section)
  Term: Second semester

- **Course:** Fundamentals of Computer Systems
  Degree Program: Bachelor's Degree in Telecommunications Technologies and Services Engineering
  Hours: 15h (1 lab section)
  Term: Second semester

Academic Years: 2020–2021, 2021–2022

- **Course:** Programming II
  Degree Program: Bachelor's Degree in Telecommunications Technologies and Services Engineering
  Hours: 60h (2 lab sections)
  Term: Second semester

Academic Years: 2022–2023, 2023–2024

- **Course:** Automata and Formal Languages
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (2 lab sections)
  Term: First semester

Academic Years: 2024–2025, 2025–2026

- **Course:** Operating Systems
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (2 lab sections)
  Term: Second semester

Academic Year: 2024–2025

- **Course:** Project Administration and Management
  Degree Program: Bachelor's Degree in Biomedical Engineering
  Hours: 30h (1 lecture section)
  Term: Second semester

Academic Years: 2024–2025, 2025–2026

- **Course:** Mobile Application Development
  Degree Program: Bachelor's Degree in Computer Engineering
  Hours: 60h (project-based course)
  Term: Second semester

Teaching experience (continued)

Course: Deep Learning Fundamentals and Basic Tools
Degree Program: Master's Degree in Deep Learning for Audio and Video Processing
Hours: 12h
Term: First semester

Short Courses and Workshops

March 4 and 11, 2024

Course: Free Software for Research Careers
Autonomous University of Madrid (UAM)

June 2024

Course: AI Workshop for the "I Want to Be an Engineer" Program
Autonomous University of Madrid (UAM)